The induction principle of n-truncation

The Univalent Foundations Program

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Editorial note. The counted statement is the author's 'thm:truncn-ind' (source lines 514-517, Theorem 7.3.2); the proof environment starts on the next line, so the pairing is by adjacency.

Editorial note. The retained context is the author's own sentence explaining that the induction principle extracted from the constructors is being reformulated as this theorem.

Editorial note. The retained counter values (chapter 7, section 3, theorem offset 1) make the retained theorem print as Theorem 7.3.2, matching the original book numbering.

Editorial note. No retained line refers to a label defined outside the retained span, so the package declares no external reference and no bibliography.

Editorial scope. One original result of the HoTT Book, counted once: the induction principle of n-truncation (Theorem 7.3.2).

Allowed standard prerequisites (editorial).

- The n-truncation as a higher inductive type. trunc n A is generated by |a|_n for a : A together with, for each r : S^{n+1} -> trunc n A and r' : prd x P(r(x)), a point h(r) and a dependent path s_r(x).. The reformulation at source lines 519-521 works directly with these extra constructors.
- Contractibility of n-types with a point. An n-type together with a point u is contractible at u: there is a contraction prd x (t x = u).. It produces u and v at source line 523, which are the second and third data of the induction principle.
- Dependent paths. dpath P p u v unfolds to trans P p u = v.. It is what turns the contraction v into the third datum at source line 525.

To make this more useful, we reformulate it as follows.

Theorem 7.3.2. *For any type family $P : \|A\|_n \to \mathcal{U}$ such that each $P(x)$ is an n-type, and any function $g : \prod_{(a:A)} P(|a|_n)$, there exists a section $f : \prod_{(x:\|A\|_n)} P(x)$ such that $f(|a|_n) :\equiv g(a)$ for all $a : A$.*

Proof. It will suffice to construct the second and third data listed above, since g has exactly the type of the first datum. Given $r : \mathbb{S}^{n+1} \to \|A\|_n$ and $r' : \prod_{(x:\mathbb{S}^{n+1})} P(r(x))$, we have $h(r) : \|A\|_n$ and $s_r : \prod_{(x:\mathbb{S}^{n+1})}(r(x) = h(r))$. Define $t : \mathbb{S}^{n+1} \to P(h(r))$ by $t(x) :\equiv s_r(x)_*(r'(x))$. Then since $P(h(r))$ is n-truncated,

there exists a point $u : P(h(r))$ and a contraction $v : \prod_{(x:\mathbb{S}^{n+1})}(t(x) = u)$. Define $h'(r, r') :\equiv u$, giving the second datum. Then (recalling the definition of dependent paths), v has exactly the type required of the third datum. $\square$